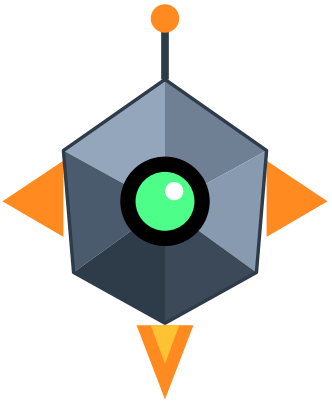
BOOK 7

The Choice Before the Calculation

Choosing your tools before you start





MULTIX//HELLO
I am Multix. I ride along in every chapter. Watch my readout: it shows what is owed, what is paid, and what is only a guess.

THE CHOICE BEFORE THE CALCULATION

CHAPTER 1

Pick Your Tools First



MULTIX // CIRCLE CHECK
A circle holds a piece that belongs together, and circles can sit inside circles. A circle is *ready* when everything needed for its calculation is known and its operation is allowed by the current rules. To **pop** a ready circle, replace it with its value and leave everything outside it alone. To **redraw**, move circles around or split a number, as long as the value stays the same. The full rules are in the introduction.

STORY

Rina had a toolbox, and it was not full of hammers. It was full of *moves* for numbers: splitting a number into smaller pieces, giving a mystery a name, and choosing a good starting point.
Her grandfather, who fixed furniture, liked to say: "The job gets easy or hard before you pick up the first tool. It depends which tool you pick."
Rina found out he was right about arithmetic too.

STEP 1 The same problem, two ways

Rina had to find **25 × 36**.

Way 1: Split the 36 into 30 and 6.

25 × 30 = 750. 25 × 6 = 150. 750 + 150 = **900**.

Way 2: Notice that 36 is 4 × 9, so 25 × 36 is 25 × 4 × 9. Now look at 25 × 4. It is 100, and 100 × 9 is **900**.

Both ways give 900. The answers match, because splitting a number into pieces never changes what it is. But Way 2 turned the hard part into a number Rina already knew, 25 × 4 = 100, so there was less to do.

Rina had not started calculating yet. The big decision, how to split 36, came *before* any arithmetic.

IN CIRCLES

Rina's choice is a choice about where to draw circles. Both of these are redraws that keep the value of (25 × 36):

Trust check. A choice is trustworthy when it is an **exact rewriting**: when the new form is the old one, just arranged differently.

TRACE

```
( 25 × 36 )
redraw  →  ( 25 × ( 4 × 9 ) )
redraw  →  ( ( 25 × 4 ) × 9 )
pop ( 25 × 4 )  →  100
( 100 × 9 )
pop  →  900
```

TRACE

```
( 25 × 36 )
redraw  →  ( 25 × ( 30 + 6 ) )
redraw  →  ( ( 25 × 30 ) + ( 25 × 6 ) )
pop ( 25 × 30 )  →  750
pop ( 25 × 6 )  →  150
( 750 + 150 )
pop  →  900
```

Both end at 900. The first draws a circle, (25 × 4), that Rina already knows how to pop. That is why it is the better choice. The answer is not different, but the circles are easier to pop.

STEP 2 Choose a good starting point

Next problem: **99 × 17**.

Rina did not want to multiply by 99. But 99 is just one less than 100, and 100 is a lovely number to multiply by. So she chose 100 as her **reference**.

100 × 17 = 1,700. That is one 17 too many, because 99 is one less than 100, so take one 17 away.

1,700 − 17 = **1,683**.

In circles, Rina's move looks like this:

TRACE

```
( 99 × 17 )
redraw  →  ( ( 100 − 1 ) × 17 )
redraw  →  ( ( 100 × 17 ) − ( 1 × 17 ) )
pop ( 100 × 17 )  →  1,700
pop ( 1 × 17 )  →  17
( 1,700 − 17 )
pop  →  1,683
```

The circle (1 × 17) is the leftover that Rina has to put back.

Here is another one. Add **17 + 18 + 19 + 20 + 21**. Instead of adding them one at a time, Rina picked the middle number, 19, as her reference. Each number is a little more or a little less than 19:

−2, −1, 0, +1, +2

Those add to zero, because the pluses and minuses cancel.

So all that is left is five 19s: 5 × 19 = **95**.

Check by adding the long way: 17 + 18 + 19 + 20 + 21 is 95. It matches.

STEP 3 Name the mystery

Some problems have a mystery number in the middle of them. Here is one.

A shirt costs 5 more than a hat. Together they cost 25. How much is the hat?

Rina gave the hat a name: **h**. Then the shirt is h + 5. Together:

h + (h + 5) = 25 2 × h + 5 = 25 2 × h = 20 h = 10

The hat costs 10, so the shirt costs 15. Check: 10 + 15 is 25.

Naming the mystery was a choice too. Rina could have guessed and tested numbers for a long time. Giving the hat a name turned the problem into something she could rearrange.

STEP 4 Choices can be good or bad, but not wrong

A choice does not change the answer. It only changes how much work you do.

For 99 × 17, Rina could have picked 90 as her reference instead of 100. Then she would work out 90 × 17 = 1,530 and 9 × 17 = 153, and add them: 1,530 + 153 = **1,683**.

Same answer. But she had two multiplications and an addition, instead of one multiplication and one subtraction. The reference 100 was a better choice for this problem.

There is one thing to be careful about. A choice stays safe only if you **put back everything you moved**. If Rina rounded 99 up to 100, got 1,700, and stopped, she would be 17 too high. That is a wrong answer, and the fix is the correction from A Useful Wrong Answer: name what you changed and take it back.

STEP 5 The note you leave

Rina's grandfather wrote a note on every repair: *"Replaced the leg, because the old one was cracked."* The note did not say how he *knew* to look at the leg. It said what he chose and why.

A good number notebook does the same. It writes down the choice:

"I split 36 into 4 × 9, because 25 × 4 is 100."

The note does not have to explain how Rina thought of it. Sometimes a good idea arrives as a hunch, which is what Kofi's

messy notebook was for. But writing the choice down means somebody else can check it.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What stayed the same when Rina changed tools?** (The answer.)
- **What changed?** (How many steps there were, and which steps were easy.)
- **What would have been lost if Rina rounded 99 up to 100 and stopped?** (The right answer. She would be 17 too high.)

WHAT MAKES THIS CHOICE TRUSTWORTHY?

A choice is trustworthy when it is an **exact rewriting**: when the new form is the old one, just arranged differently. 36 really is 4×9. 99 really is $100 - 1$. A hat plus 5 really is the shirt.
A choice stops being trustworthy when something is changed and not put back, such as rounding 99 to 100 and forgetting the leftover.

PRACTICE

1. Find **25 × 48** by splitting 48 so that you get a 25 × 4 inside it.
2. Find **98 × 12** by choosing 100 as your reference. What do you have to take back?
3. Find **15 + 16 + 17 + 18 + 19** by choosing the middle number as your reference.
4. A notebook costs 3 more than a pencil. Together they cost 11. Name the pencil p, and find both prices.
5. Find **125 × 16**. (Hint: what is 125 × 8?)
6. Find **45 × 22** in two different ways. Which way took fewer steps for you?
7. Write **25 × 28** with circles so that (25 × 4) is ready to pop. Then pop everything.

BRIDGE *Bridge to the rest of the series:* In The Work That Vanishes, choosing 50 as the middle number is a choice of reference, and it is what made the corner appear. In What Must Remain Unresolved, naming the jar x is the same move as naming the hat h. In Two Boxes, One Answer, two different choices give the same answer, so they look identical from the slot, though one took much more work inside.

PUZZLE PAGE

Color by Answer

Shade every box whose answer is exactly **2,400**. Leave the others blank. What appears?

8 × 299	2384 + 16	2414 − 14	2370 + 20	2370 + 28	55 × 55 − 25 × 25	2421 − 21	2396 + 16
2379 + 21	2419 − 19	2377 + 23	2411 − 11	2419 − 19	2419 − 19	125 × 125 − 115 × 115	2385 + 15
2371 + 29	2428 − 28	2364 + 36	2368 + 32	2422 − 22	2397 + 3	125 × 125 − 115 × 115	2390 + 10
2406 − 6	2395 + 5	2391 + 9	2382 + 18	2388 + 12	2363 + 37	70 × 70 − 50 × 50	50 × 48
2410 + 2	2381 + 19	2369 + 31	2420 − 20	2417 − 17	2422 − 22	16 × 150	2394 + 4
2401 + 3	2387 + 7	2408 − 8	2378 + 22	24 × 100	2416 − 16	2376 + 26	2387 + 12
2376 + 19	14 × 172	104 × 23	2437 − 37	2433 − 33	2365 + 25	51 × 47	51 × 51 − 14 × 14
76 × 76 − 58 × 58	3 × 796	6 × 402	2383 + 11	2379 + 11	2407 + 5	2391 + 15	2387 + 14



MULTIX//SCAN

Some boxes look close and are not. Check the corner on each one.

THE CHOICE BEFORE THE CALCULATION

CHAPTER 2

Where Do You Start Counting?

STORY

Rina had measured how many steps she walked to school on four mornings: 47, 38, 52 and 41. She wanted the total number of steps.
She could add them directly. But Rina had a habit of choosing before calculating, so she asked a different question: *is there a number near all four that I can start from?*

STEP 1 Pick a starting number

All four numbers are near 45. Rina decided to start at 45, and for each day write how far it is above or below 45.

Day	Steps	Compared with 45
1	47	2 above
2	38	7 below
3	52	7 above
4	41	4 below

Now the total is four lots of 45, with a few small corrections. Here it is in circles:

TRACE

```
( 47 + 38 + 52 + 41 )
redraw  →  ( ( 45 + 2 ) + ( 45 − 7 ) + ( 45 + 7 ) + ( 45 − 4 ) )
redraw  →  ( ( 4 × 45 ) + 2 + 7 − 7 − 4 )
```

She grouped the corrections. Above: 2 + 7 = 9. Below: 7 + 4 = 11. The corrections are 9 up and 11 down, which is 2 down overall.

TRACE

```
( 4 × 45 )  →  180
( 11 − 9 )  →  2
( 180 − 2 )  →  178
```

The total is **178** steps. Check by adding directly: 47 + 38 = 85, 85 + 52 = 137, 137 + 41 = 178. ✓

Trust check. The reason is that each redraw keeps the value.

STEP 2 Does the start change the answer?

Rina tried two other starting points to see.

Start at 40. The days compare as 7 above, 2 below, 12 above, 1 above. Above: 7 + 12 + 1 = 20. Below: 2. Net: 18 above.

TRACE

```
( ( 4 × 40 ) + ( 20 − 2 ) )  →  ( 160 + 18 )  →  178
```

Start at 100. Every day is below 100: by 53, 62, 48 and 59. Together that is 53 + 62 + 48 + 59 = 222 below.

TRACE

```
( ( 4 × 100 ) − 222 )  →  ( 400 − 222 )  →  178
```

All three starts give **178**. The answer does not depend on where Rina starts counting.

STEP 3 But some starts are kinder than others

Starting at 45 left corrections of 2, 7, 7 and 4. Easy. Starting at 40 left 7, 2, 12 and 1. Fine. Starting at 100 left 53, 62, 48 and 59. Those are big numbers to add, and a slip is more likely.

So the choice matters for how hard the work is and how likely you are to slip, but not for the answer. A good starting point is a number that is **close to everything** and **easy to multiply by**. The number 45 was close, and 4 × 45 was easy. A number like 44 would have been close too, but 4 × 44 is less friendly.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What stayed the same with every start?** (The total, 178.)
- **What changed with the start?** (The size of the corrections, and so how hard the work was.)

WHAT MAKES THIS TRUSTWORTHY?

The reason is that each redraw keeps the value. A number like 47 is exactly 45 + 2, so writing it that way loses nothing. Any start works for the same reason.
The check is to add the numbers directly, or to try a second start. If two different starts agree, a slip is much less likely, because two different slips rarely land on the same wrong number.

PRACTICE

1. Add 98 + 103 + 101 + 99 using a start of 100. Show the corrections.
2. Add 52 + 48 + 51 + 49 + 50 using a start of 50.
3. Add 47 + 38 + 52 + 41 using a start of 50. Do you get 178?
4. For 62 + 58 + 61 + 59, which start is kind? Why?
5. Rina says “starting at 1,000 would also work for 47 + 38 + 52 + 41.” Is she right? Is it a good choice?

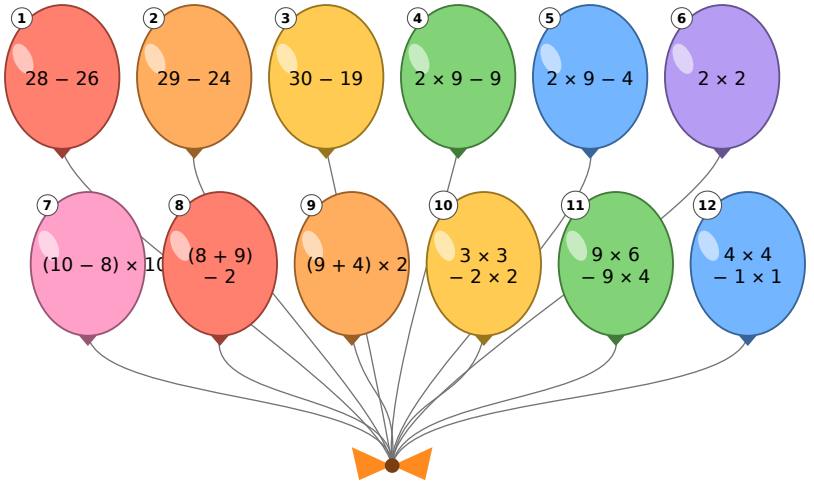
BRIDGE *Bridge to the rest of the series:* In chapter 1, Rina picked how to split a number. Here she picks where to start. Both choices leave the answer alone and change how hard the work is. In A Useful Wrong Answer, the corrections were small because the estimate was near.

PUZZLE PAGE

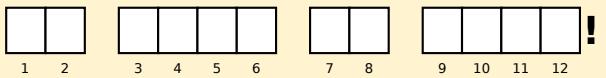
The Balloon Message

What is Multix's advice? Work out each balloon, use the key to turn the number into a letter, and fill the banner box with the same number.

A = 1	B = 2	C = 3	D = 4	E = 5	F = 6	G = 7	H = 8	I = 9	J = 10	K = 11	L = 12	M = 13
N = 14	O = 15	P = 16	Q = 17	R = 18	S = 19	T = 20	U = 21	V = 22	W = 23	X = 24	Y = 25	Z = 26



What was the answer?



CHAPTER 3

Percent Is Just a Friendly Piece

STORY

A sign in the shop window said **15% off**. The jacket cost 80 dollars. Rina wanted to know how much she would save.
“15 percent of 80,” said Rina. “That sounds hard. But I get to choose how to cut it.”

STEP 1 Cut the percent into friendly pieces

Some percents are easy. **10%** of a number is that number divided by 10. **5%** is half of 10%. **1%** is the number divided by 100.

15 is 10 plus 5. So 15% of 80 is 10% of 80 plus 5% of 80.

TRACE

```
( 15% of 80 )
redraw  →  ( ( 10% of 80 ) + ( 5% of 80 ) )
pop  →  ( 8 + 4 )
pop  →  12
```

Rina saves **12 dollars**. The jacket costs 80 − 12 = **68 dollars**.

STEP 2 Different percents, different pieces

Rina could pick pieces that fit the percent she had.

35% of 60. Use 30% and 5%. 10% of 60 is 6, so 30% is 18, and 5% is 3.

TRACE

```
( 35% of 60 )  →  ( 18 + 3 )  →  21
```

25% of 48. 25% is a quarter, so divide by 4.

TRACE

```
( 25% of 48 )  →  12
```

99% of 200. 99 is almost 100. 100% of 200 is 200, and the 1% that is missing is 2.

Trust check. The reason is that percents add.

TRACE

```
( 99% of 200 )  →  ( 200 − 2 )  →  198
```

Look at the pattern. Rina did not run one method on every problem. She looked at the percent, then chose the friendly piece that fits.

STEP 3 Name the thing you want

One more choice: **what do you want to find?** Sometimes Rina knows the saving and wants the price, and sometimes it is the other way round.

The jacket saves 12 dollars out of 80. To find the price she pays, she can either:

- subtract the saving: 80 − 12 = 68, or
- notice that paying 85% means 10% plus 5% less than the whole: 80 − 8 − 4 = 68.

Both give 68. Pick the one that matches the question. If the sign said “pay 85%”, Rina would use the second. If the sign said “save 15%”, the first.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What stayed the same after we cut 15% into 10% and 5%?** (The value, 12.)
- **What changed?** (The work: two easy steps instead of one hard one.)

WHAT MAKES THIS TRUSTWORTHY?

The reason is that percents add. 10% of a number plus 5% of the same number is 15% of that number, because both are parts of the same whole. So the redraw keeps the value.
The check is to try a different cut. For 15% of 80, use 20% minus 5%: 16 − 4 = 12. ✓ Two different cuts that agree make a slip unlikely.

PRACTICE

1. Find 15% of 60.
2. Find 20% of 35.
3. Find 45% of 40. (Hint: 40% and 5%.)

4. Find 98% of 50.

5. A bike costs 120 dollars and is 15% off. What does it cost after the discount?

6. Rina finds 15% of 80 as 20% minus 5%. Show her notebook in circles.

BRIDGE *Bridge to the rest of the series:* In chapter 2, Rina chose a start. Here she chooses friendly pieces of a percent. Both choices change the work and leave the answer alone. In The Work That Vanishes, $99 \times 101 = 100 \times 100 - 1$ uses the same idea of leaning on a friendly number.

PUZZLE PAGE

Spot the Slip

Three notebooks. One is perfect. The other two each hide a single slip. Find it!

TRACE

```
1 ( 37 × 43 )
2 redraw  →  ( ( 40 − 3 ) × ( 40 + 3 ) )
3 redraw  →  ( ( 40 × 40 ) − ( 3 × ( 3 + 1 ) ) )
4 pop ( 40 × 40 )  →  1,600
5 pop ( 3 × ( 3 + 1 ) )  →  12
6 pop ( 1,600 − 12 )  →  1,588
```

Notebook A ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 19 × 21 )
2 redraw  →  ( ( 20 − 1 ) × ( 20 + 1 ) )
3 redraw  →  ( ( 20 × 20 ) − ( 1 × 1 ) )
4 pop ( 20 × 20 )  →  400
5 pop ( 1 × 1 )  →  2
6 pop ( 400 − 2 )  →  398
```

Notebook B ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 64 × 76 )
2 redraw  →  ( ( 70 − 6 ) × ( 70 + 6 ) )
3 redraw  →  ( ( 70 × 70 ) − ( 6 × 6 ) )
4 pop ( 70 × 70 )  →  4,900
5 pop ( 6 × 6 )  →  36
6 pop ( 4,900 − 36 )  →  4,864
```

Notebook C ☐ **perfect** ☐ **slip on line** _____



MULTIX // HINT

A slip can hide in a **pop** (wrong value) or in a **redraw** (value changed). Check each line against the problem at the top.

THE CHOICE BEFORE THE CALCULATION

CHAPTER 4

Start From the Extreme

STORY

Rina's piggy bank held 12 coins, all nickels and dimes. Together they were worth 85 cents. How many of each? She could guess and check for ages. Instead she chose where to start.

STEP 1 Pretend everything is the smaller coin

"Suppose all 12 coins were nickels," said Rina.

TRACE

```
( 12 × 5 )  →  60
```

That is only 60 cents, but the bank holds 85. She is 25 cents short.

STEP 2 Swap coins one at a time

If Rina swaps one nickel for a dime, the number of coins stays 12, and the total goes up by 5 cents (a dime is 10, a nickel is 5).

How many swaps close the gap of 25?

TRACE

```
( 25 ÷ 5 )  →  5
```

Five swaps. So there are **5 dimes** and 12 − 5 = **7 nickels**.

Trust check. The reason is that each swap does the same thing: it adds a fixed amount (5 cents, or 2 legs) and leaves the count alone.

STEP 3 Check it

TRACE

```
( ( 7 × 5 ) + ( 5 × 10 ) )
pop  →  ( 35 + 50 )
pop  →  85
```

85 cents, and 7 + 5 = 12 coins. ✓

Rina did not guess. She picked a starting point that was easy to work out (all nickels), looked at how far off she was, and found what one swap did.

STEP 4 The same idea on another problem

A farmyard has 10 animals, all chickens and cows. Together they have 28 legs. How many of each?

Start with all chickens (2 legs each):

TRACE

```
( 10 × 2 )  →  20
```

That is 8 legs short. Swapping a chicken for a cow adds 2 legs. So (8 ÷ 2) = 4 swaps. There are **4 cows** and **6 chickens**.

Check: ((6 × 2) + (4 × 4)) = 12 + 16 = 28. ✓

The choice was to start at the extreme, where every animal was the same kind, and then fix the gap.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the extreme start keep?** (The number of coins or animals.)
- **What did it change?** (The total value or legs, which we fixed with swaps.)

WHAT MAKES THIS TRUSTWORTHY?

The reason is that each swap does the same thing: it adds a fixed amount (5 cents, or 2 legs) and leaves the count alone. So the gap divided by the amount per swap is exactly the number of swaps.
The check, as always, is to take the answer back to the original: 7 nickels and 5 dimes really do make 12 coins and 85 cents.

PRACTICE

1. There are 9 coins, all nickels and dimes, worth 70 cents. How many of each? Start with all nickels.
2. There are 15 bicycles and tricycles in a shed, with 40 wheels in total. A bicycle has 2 wheels and a tricycle has 3. How many of each?
3. There are 6 coins, nickels and dimes, worth 45 cents. How many of each?
4. Rina starts with all dimes in the piggy-bank problem instead. How much too high is the total, and how

much does one swap change it?

5. Why does the swap trick need all the coins to be one of two kinds?

BRIDGE *Bridge to the rest of the series:* In chapter 3, Rina cut a percent into friendly pieces. Here she picks a friendly extreme to start from. In Reversing the Calculation, working backward from a clue and starting at an extreme are two ways to avoid guessing.

PUZZLE PAGE

Color by Answer

Shade every box whose answer is exactly **1,225**. Leave the others blank. What appears?

1226 + 7	1233 − 8	22 × 48 + 13 × 13	45 × 27	1221 + 8	1222 + 3	35 × 35	1211 + 6
1239 − 14	7 × 175	1203 + 22	1227 − 2	1242 − 17	7 × 63 + 28 × 28	1232 − 7	1205 + 20
1204 + 21	1232 − 7	1187 + 38	1195 + 30	23 × 47 + 12 × 12	1207 + 18	1199 + 26	17 × 53 + 18 × 18
20 × 50 + 15 × 15	14 × 56 + 21 × 21	1236 − 11	1232 − 7	1213 + 12	1209 + 16	1229 − 4	1238 − 13
1210 + 23	1262 − 37	1242 − 17	91 × 91 − 84 × 84	4 × 66 + 31 × 31	18 × 52 + 17 × 17	125 × 125 − 120 × 120	1188 + 29
1222 + 6	1217 + 14	1233 − 8	1233 − 8	1232 − 7	1203 + 22	1206 + 11	1216 + 8
1229 + 2	1207 + 9	1199 + 16	35 × 35	8 × 62 + 27 × 27	1226 + 7	1208 + 10	1187 + 29
3 × 406	14 × 87	1213 + 24	1228 + 5	72 × 17	124 × 124 − 119 × 119	1208 + 5	1223 + 4



MULTIX//SCAN

Some boxes look close and are not. Check the corner on each one.

THE CHOICE BEFORE THE CALCULATION

CHAPTER 5

Name the Unknown Well

STORY

Rina's brother Leo is some age, and her cousin Mia is 3 years older. Their ages add up to 21. How old is each? Rina decided to give the unknown a name before doing anything else.

STEP 1 Name the small one

Let **L** be Leo's age. Then Mia's age is L + 3.

TRACE

```
( L + ( L + 3 ) )  =  21
```

Rina redraws the circle so that the two L's are together.

TRACE

```
( L + ( L + 3 ) )
redraw  →  ( ( L + L ) + 3 )
```

The circle is stuck on L, but she knows the total is 21. So (L + L) must be 21 − 3.

TRACE

```
( L + L )  =  ( 21 − 3 )  →  18
```

Two lots of Leo's age make 18, so Leo is **9** and Mia is **12**. Check: 9 + 12 = 21, and 12 is 3 more than 9. ✓

STEP 2 Choosing the name changes the work

Now a different puzzle: three numbers in a row, such as 5, 6, 7, add up to 48. Which three?

Rina could name the smallest, **s**: then the numbers are s, s + 1, s + 2.

TRACE

```
( s + ( s + 1 ) + ( s + 2 ) )  =  48
```

That works, but there are two extra bits to carry (the +1 and +2).

Trust check. The answer is checked by going back to the words: 9 and 12 are ages, 12 is 3 more than 9, and they add to 21.

Or she could name the **middle** one, **m**. Then the numbers are m − 1, m and m + 1.

TRACE

```
( ( m − 1 ) + m + ( m + 1 ) )
```

Now the −1 and +1 cancel each other out. What is left is three lots of m.

TRACE

```
( 3 × m )  =  48
```

So m = 48 ÷ 3 = **16**, and the numbers are **15, 16 and 17**. Check: 15 + 16 + 17 = 48. ✓

The answers are the same. But naming the middle made the extra bits vanish by themselves. That is what a good name does.

STEP 3 What makes a name good?

Rina found three habits.

- **Name the thing you want to find**, not some other quantity.
- **Say what the name stands for**, in words: "L is Leo's age," not just "L."
- **Choose the name that makes the other pieces simple.** If the others are all "plus or minus a little", the middle one often works best.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did naming the unknown keep?** (The whole problem, in a form we can work on.)
- **What did the choice of name change?** (How many extra bits had to be carried.)

WHAT MAKES THIS TRUSTWORTHY?

The answer is checked by going back to the words: 9 and 12 are ages, 12 is 3 more than 9, and they add to 21. If any of those fails, the name or the setup was wrong. A good name also makes the setup easy to read, which makes slips easier to see.

PRACTICE

1. Two numbers add to 30 and differ by 6. Name the smaller one s and find both.
2. Three whole numbers in a row add to 63. Name the middle one and find all three.
3. A pencil costs 5 dollars more than a pen. Together they cost 17 dollars. Name the pen's price p and find both prices.
4. Check your answer to problem 3 against the words.
5. Why is it better to say "L is Leo's age" than just "let L be a number"?

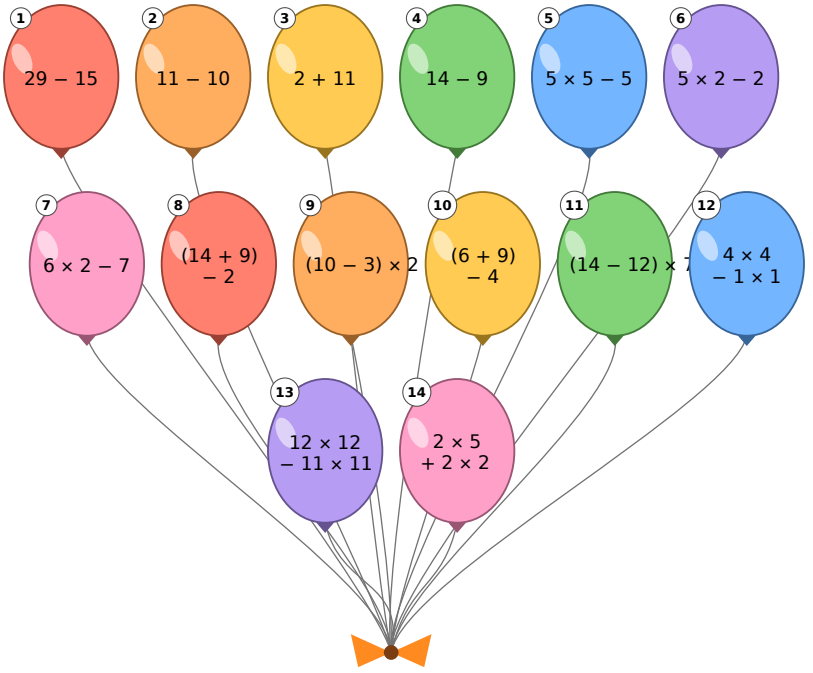
BRIDGE *Bridge to the rest of the series:* In What Must Remain Unresolved, an unknown is a number that stays a number until it is known. Here the choice of name changes the work, just as the choice of tools did in chapter 1. In The Work That Vanishes, the -1 and $+1$ that cancel are a small version of the missing corner.

PUZZLE PAGE

The Balloon Message

What is Multix's advice? Work out each balloon, use the key to turn the number into a letter, and fill the banner box with the same number.

A = 1	B = 2	C = 3	D = 4	E = 5	F = 6	G = 7	H = 8	I = 9	J = 10	K = 11	L = 12	M = 13
N = 14	O = 15	P = 16	Q = 17	R = 18	S = 19	T = 20	U = 21	V = 22	W = 23	X = 24	Y = 25	Z = 26



What was the answer?

													!
1	2	3	4	5	6	7	8	9	10	11	12	13	14

THE CHOICE BEFORE THE CALCULATION

CHAPTER 6

Choose the Unit

STORY

Rina’s trips to the pool take 45 minutes each. She makes 4 trips a week. How long does she spend altogether? She could calculate in hours or in minutes. She stopped to choose.

STEP 1 Two units, two calculations

In hours. 45 minutes is 0.75 hours.

TRACE

```
( 4 × 0.75 )  →  3
```

In minutes. Stay with 45.

TRACE

```
( 4 × 45 )  →  180
```

Then 180 minutes is 3 hours. Both give the same answer, 3 hours. The minutes version used whole numbers all the way, which many people find easier. The hours version needed a decimal.

STEP 2 Money in cents

Three notebooks cost 1.99 dollars each.

In dollars. (3 × 1.99) has a decimal that is awkward.

In cents. 1.99 dollars is 199 cents.

TRACE

```
( 3 × 199 )
redraw  →  ( ( 3 × 200 ) − ( 3 × 1 ) )
pop  →  ( 600 − 3 )
pop  →  597
```

597 cents is **5.97 dollars**.

Choosing cents turned a decimal problem into a whole-number problem, and then a friendly number (200) helped with the correction.

Trust check. The reason is that a unit is only a size of measure, and converting is exact: 45 minutes is 0.75 hours, and 199 cents is 1.99 dollars.

STEP 3 Mixed units

Rina walks for 2.5 hours and then rests for 90 minutes. How long is that in total?

The units do not match, so she picks one.

TRACE

```
2.5 hours  =  150 minutes
( 150 + 90 )  →  240 minutes  =  4 hours
```

If she had added 2.5 and 90 directly, the answer would be meaningless, because hours and minutes are different sizes. First choose the unit, then calculate.

STEP 4 What makes a unit good?

Rina's rules:

- Pick the unit that makes **all** the numbers whole, if you can.
- Make sure **all** the quantities use the **same** unit before adding or subtracting them.
- Convert back at the end, so the answer is in a unit people expect.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did changing the unit keep?** (The real amount of time or money.)
- **What changed?** (The numbers we wrote, and how easy the arithmetic was.)

WHAT MAKES THIS TRUSTWORTHY?

The reason is that a unit is only a size of measure, and converting is exact: 45 minutes is 0.75 hours, and 199 cents is 1.99 dollars. The answer in either unit describes the same thing.
The check is to convert the answer back and compare: 5.97 dollars is 597 cents, and 3 hours is 180 minutes. ✓

PRACTICE

1. Six trips of 15 minutes each. How many minutes? How many hours?
2. Five notebooks at 2.98 dollars each. Work in cents

and give the total in dollars.

3. Rina cycles 1.25 hours and then walks 45 minutes. What is the total in minutes? In hours?

4. How many hours are in 3 days and 12 hours?

5. Why is it wrong to add 2.5 hours and 90 minutes as 2.5 + 90?

BRIDGE *Bridge to the rest of the series:* In chapter 3, Rina chose friendly percent pieces. Here she chooses a friendly unit. In A Useful Wrong Answer, mixing up units is one more way for an answer to be wrong by a lot.

PUZZLE PAGE

Spot the Slip

Three notebooks. One is perfect. The other two each hide a single slip. Find it!

TRACE

```
1 ( 38 × 42 )
2 redraw  →  ( ( 40 − 2 ) × ( 40 + 2 ) )
3 redraw  →  ( ( 40 × 40 ) − ( 2 × 2 ) )
4 pop ( 40 × 40 )  →  1,600
5 pop ( 2 × 2 )  →  4
6 pop ( 1,600 − 4 )  →  1,596
```

Notebook A ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 24 × 26 )
2 redraw  →  ( ( 25 − 1 ) × ( 25 + 1 ) )
3 redraw  →  ( ( 25 × 25 ) − ( 1 × 1 ) )
4 pop ( 25 × 25 )  →  625
5 pop ( 1 × 1 )  →  2
6 pop ( 625 − 2 )  →  623
```

Notebook B ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 46 × 54 )
2 redraw  →  ( ( 50 − 4 ) × ( 50 + 4 ) )
3 redraw  →  ( ( 50 × 50 ) − ( 4 × ( 4 + 1 ) ) )
4 pop ( 50 × 50 )  →  2,500
5 pop ( 4 × ( 4 + 1 ) )  →  20
6 pop ( 2,500 − 20 )  →  2,480
```

Notebook C ☐ **perfect** ☐ **slip on line** _____



MULTIX // HINT

A slip can hide in a **pop** (wrong value) or in a **redraw** (value changed). Check each line against the problem at the top.

THE CHOICE BEFORE THE CALCULATION

CHAPTER 7

Pick a Picture

STORY

Rina had 3 shirts and 4 hats. How many different outfits, with one shirt and one hat, can she make?
She could list them in a messy row. Instead she chose a picture.

STEP 1 A table

Rina drew a table. Rows are shirts, columns are hats, and each box is one outfit.

	Hat 1	Hat 2	Hat 3	Hat 4
Shirt 1	✓	✓	✓	✓
Shirt 2	✓	✓	✓	✓
Shirt 3	✓	✓	✓	✓

Three rows of four boxes each.

TRACE

```
( 3 × 4 )  →  12
```

There are **12** outfits. The picture made it plain why the answer is a product: 3 rows of 4.

STEP 2 When the picture shows a rule

Now Rina says: *the red shirt must never go with the green hat.* Shirt 1 is red and Hat 4 is green.

The table shows that exactly one box is not allowed. She crosses it out:

	Hat 1	Hat 2	Hat 3	Hat 4
Shirt 1	✓	✓	✓	✗
Shirt 2	✓	✓	✓	✓
Shirt 3	✓	✓	✓	✓

Trust check. A picture is trustworthy when every item in the problem appears in it exactly once.

TRACE

```
( 12 − 1 )  →  11
```

There are **11** outfits that are allowed.

This is the idea from The Boundary of a Step: the table lists every outfit that is **possible**, and the rule says which are **permitted**. Counting the permitted ones means counting all and taking away the refused ones, as in Count the Other Side.

STEP 3 A bar for fractions

Rina has 24 stickers and gives away three quarters of them. How many is that?

She draws a bar of 24 and cuts it into four equal parts.

TRACE

```
| 6 | 6 | 6 | 6 |
```

Each part is 24 ÷ 4 = 6. Three quarters is three parts.

TRACE

```
( 3 × 6 )  →  18
```

She gives away **18** stickers and keeps 6.

Here the picture turned a fraction into something Rina could see and count.

STEP 4 Choosing which picture

Different questions suggest different pictures.

Question	Good picture
How many pairs (one of this, one of that)?	a table
What part of a total?	a bar
How far apart?	a number line
How many in rows and columns?	a grid

The picture does not change the answer. It changes how easy the answer is to see, and how easy it is to spot a rule.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the table keep?** (Every outfit, one per box.)
- **What did it make easy?** (Seeing the product, and seeing the refused outfit.)

WHAT MAKES THIS TRUSTWORTHY?

A picture is trustworthy when every item in the problem appears in it exactly once. In the table, each box is one outfit, and none is missing or counted twice. The check is to count another way: 4 hats for each of 3 shirts is also 12.

PRACTICE

1. Rina has 5 shirts and 3 hats. How many outfits? Draw the table in your head and say the product.
2. She cannot wear shirt 2 with hat 1 or hat 2. How many outfits are allowed?
3. She gives away two thirds of 30 stickers. Use a bar.
4. A grid has 6 rows and 7 columns. How many boxes? Which picture is it?
5. Why is a table better than a list for the outfit problem?

BRIDGE *Bridge to the rest of the series:* In chapter 1, Rina picked a way to split a number. Here she picks a picture before counting. In Solving by Changing the Question, counting the refused ones and subtracting was a way to change the question, and here the table shows it.
Next in this book: Chapter 8: Why Split at Fifty.

PUZZLE PAGE

Color by Answer

Shade every box whose answer is exactly **899**. Leave the others blank. What appears?

13 × 69	937 − 38	902 − 3	886 + 8	902 + 5	928 − 29	919 − 20	894 + 9
29 × 31	924 − 25	927 − 28	931 − 32	896 + 3	871 + 28	922 − 23	919 − 20
862 + 37	865 + 34	892 + 7	915 − 16	869 + 30	936 − 37	892 + 7	916 − 17
892 + 7	918 − 19	916 − 17	887 + 12	930 − 31	911 − 12	907 − 8	938 − 39
875 + 29	923 − 24	916 − 17	869 + 30	912 − 13	918 − 19	867 + 32	884 + 25
885 + 21	881 + 19	938 − 39	924 − 25	879 + 20	30 × 30 − 1 × 1	884 + 17	892 + 5
895 + 14	879 + 10	877 + 15	936 − 37	894 + 5	11 × 82	879 + 22	887 + 19
876 + 15	884 + 3	882 + 13	30 × 30 − 3 × 3	882 + 26	877 + 13	861 + 27	866 + 25



MULTIX // SCAN

Some boxes look close and are not. Check the corner on each one.

CHAPTER 8

Why Split at Fifty

STORY

Priya had the case file card **48 × 52** and a free choice. The calculation had not started. She could split the numbers in many ways, and the way she picked would decide how hard the rest was.

STEP 1 Two splits, side by side

Split 1: take 48 apart as 40 + 8.

TRACE

```
( 48 × 52 )
redraw  →  ( ( 40 × 52 ) + ( 8 × 52 ) )
```

Both circles are ready, but they are not friendly. 40 × 52 and 8 × 52 are each a small calculation of their own. Priya would probably need to split again before she could pop either one.

Split 2: take both numbers apart around 50.

TRACE

```
( 48 × 52 )
redraw  →  ( ( 50 − 2 ) × ( 50 + 2 ) )
redraw  →  ( ( 50 × 50 ) − ( 2 × 2 ) )
```

Both circles are friendly. 50 × 50 and 2 × 2 can be popped in your head.

Both splits are correct. They give the same answer, 2,496. The choice was about how friendly the pops would be.

STEP 2 What makes a split friendly?

Priya made a short list of what she looks for before she starts.

- **A friendly number near both.** 50 is near 48 and near 52, so both can move to it.
- **Matching distances.** 2 below and 2 above means the pieces will cancel, as you saw in The Work That Vanishes.
- **Pops you can do at once.** 50 × 50 and 2 × 2 need no further splitting.

Trust check. The choice cannot make the answer wrong, because every legal redraw keeps the value.

STEP 3 When there is no number near both

Priya tried **47 × 52**. The two numbers are 5 apart, so the middle is 49 and a half, which is not friendly. There is no whole number with matching distances.

She did not give up. She picked a friendly number near *one* of them.

TRACE

```
Choice A:  ( 47 × 52 )  →  ( ( 47 × 50 ) + ( 47 × 2 ) )   =  2,350 + 94    =  2,444
Choice B:  ( 47 × 52 )  →  ( ( 50 × 52 ) − ( 3 × 52 ) )   =  2,600 − 156   =  2,444
```

Both routes reach 2,444. Neither has matching distances, so the pieces do not cancel, but each round number makes one of the pops easy.

So the rule is not “always split at fifty”. It is: **look for a friendly number near both. If there is none, look for one near either.**

STEP 4 Choosing is part of the trace

Priya writes the reason for her choice in the notebook, next to the first redraw: *“around 50, matching distances.”* Later, she can see what she chose and why, and if the pops turn out hard, she can try another split.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did every split keep?** (The value, because each was a legal redraw.)
- **What changed with the choice?** (How friendly the later pops were.)

WHAT MAKES THIS TRUSTWORTHY?

The choice cannot make the answer wrong, because every legal redraw keeps the value. What it changes is cost and the chance of a slip. A good choice is one you can explain: Priya can say *why* 50 was a good place to split.

PRACTICE

1. Which split is friendlier for **99 × 101**: both numbers around 100, or 99 taken apart as 90 + 9? Find the answer.
2. Find **21 × 19** by splitting around a friendly number.

Say which number you chose.

3. Find **63 × 57** by splitting around 60.

4. For **52 × 53**, there is no whole number with matching distances. Pick a split with a friendly number near one of them and find the answer.

5. Priya says, “I chose a bad split, so my answer is wrong.” Is she right?

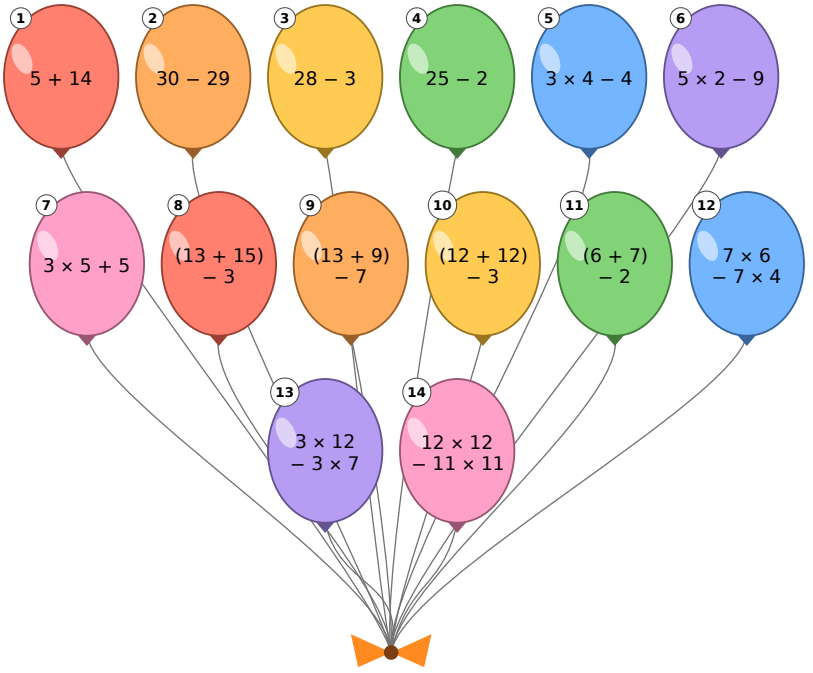
BRIDGE *Bridge to the rest of the series:* In The Boundary of a Step, the cost of the idea and the cost of a check are counted. Splitting is one more cost to weigh. In Solving by Changing the Question, a split at a friendly number is one way of changing the question, and the price is the corner.

PUZZLE PAGE

The Balloon Message

What is Multix's advice? Work out each balloon, use the key to turn the number into a letter, and fill the banner box with the same number.

A = 1	B = 2	C = 3	D = 4	E = 5	F = 6	G = 7	H = 8	I = 9	J = 10	K = 11	L = 12	M = 13
N = 14	O = 15	P = 16	Q = 17	R = 18	S = 19	T = 20	U = 21	V = 22	W = 23	X = 24	Y = 25	Z = 26



What was the answer?

													!
1	2	3	4	5	6	7	8	9	10	11	12	13	14

Answers (turn the book around)

Chapter 1

1. 48 is 4 × 12, so 25 × 48 is 25 × 4 × 12 = 100 × 12 = **1,200**.
2. 100 × 12 = 1,200, but 98 is 2 less than 100, so you have two 12s too many. Take away 2 × 12 = 24. 1,200 − 24 = **1,176**.
3. The middle number is 17. The differences −2, −1, 0, +1, +2 add to zero, so the total is 5 × 17 = **85**.
4. The notebook is p + 3. So p + (p + 3) = 11, which gives 2 × p + 3 = 11, so 2 × p = 8 and p = 4. The pencil costs **4** and the notebook costs **7**.
5. 125 × 8 = 1,000, and 16 is 8 × 2, so 125 × 16 = 1,000 × 2 = **2,000**.
6. Any of these works, and all give **990**. One way: 45 × 20 + 45 × 2 = 900 + 90. Another: 22 × 50 − 22 × 5 = 1,100 − 110. Another: 45 = 5 × 9, so 5 × 9 × 22 = 9 × 110.
7. 28 is 4 × 7, so (25 × 28) redraws as ((25 × 4) × 7). Pop (25 × 4) to get 100, and then (100 × 7) pops to **700**.

Color by Answer (2,400). The shaded boxes make a heart.



Chapter 2

1. Corrections: 2 below, 3 above, 1 above, 1 below. Above: 3 + 1 = 4. Below: 2 + 1 = 3. Net 1 above. ((4 × 100) + (4 − 3)) pops to **401**.
2. Corrections: 2 above, 2 below, 1 above, 1 below, 0. They cancel to 0. (5 × 50) pops to **250**.
3. Corrections from 50: 3 below, 12 below, 2 above, 9 below. Below: 3 + 12 + 9 = 24. Above: 2. Net 22 below. (200 − 22) pops to **178**. Yes.
4. A start of **60** is kind, because all four numbers are within 2 of it. (4 × 60) pops to 240, and the corrections 2 above, 2 below, 1 above, 1 below cancel out. Total **240**.
5. She is right, the answer would still be 178, but it is not a good choice. The corrections would be big (953, 962, 948, 959) and easy to slip on.

Balloon Message.

1.	28 − 26	= 2	B
2.	29 − 24	= 5	E
3.	30 − 19	= 11	K
4.	2 × 9 − 9	= 9	I
5.	2 × 9 − 4	= 14	N
6.	2 × 2	= 4	D
7.	(10 − 8) × 10	= 20	T
8.	(8 + 9) − 2	= 15	O
9.	(9 + 4) × 2	= 26	Z
10.	3 × 3 − 2 × 2	= 5	E
11.	9 × 6 − 9 × 4	= 18	R
12.	4 × 4 − 1 × 1	= 15	O

Message: **BE KIND TO ZERO!**

Answers (turn the book around)

Chapter 3

1. 10% is 6 and 5% is 3, so **9**.
2. 10% is 3.5, so 20% is **7**.
3. 10% of 40 is 4, so 40% is 16, and 5% is 2. 16 + 2 = **18**.
4. 100% of 50 is 50 and 2% is 1, so **49**.
5. 10% is 12 and 5% is 6, so 15% is 18. The price is 120 − 18 = **102 dollars**.
6. ((20% of 80) − (5% of 80)) → (16 − 4) → **12**.

Spot the Slip.
Notebook A: slip on line **3**. The corner is 3 × 3, not 3 × (3 + 1). The answer should be 1,591.
Notebook B: slip on line **5**. 1 × 1 is 1. The answer should be 399.
Notebook C: **perfect**. 64 × 76 = 4,864.

Chapter 4

1. All nickels: 9 × 5 = 45. The gap is 25, and each swap adds 5, so 5 swaps. That is **5 dimes and 4 nickels**. Check: 50 + 20 = 70. ✓
2. All bicycles: 15 × 2 = 30. The gap is 10, and each swap adds 1, so 10 swaps. That is **10 tricycles and 5 bicycles**. Check: 30 + 10 = 40. ✓
3. All nickels: 30. The gap is 15, and each swap adds 5, so 3 swaps. That is **3 dimes and 3 nickels**. Check: 30 + 15 = 45. ✓
4. All dimes: 12 × 10 = 120, which is 35 too high. One swap from a dime to a nickel takes 5 away. 35 ÷ 5 = 7 swaps, so 7 nickels and 5 dimes. Same answer.
5. With only two kinds, each swap changes the total by the same amount. With three kinds, the swaps differ, and the gap cannot be closed by counting equal swaps.

Color by Answer (1,225). The shaded boxes make a heart.



Answers (turn the book around)

Chapter 5

1. The numbers are s and s + 6. (s + (s + 6)) = 30, so (s + s) = 24 and s = 12. The numbers are **12 and 18**.
2. (3 × m) = 63, so m = 21. The numbers are **20, 21 and 22**.
3. (p + (p + 5)) = 17, so (p + p) = 12 and p = 6. The pen is **6 dollars** and the pencil is **11 dollars**.
4. 6 + 11 = 17, and 11 is 5 more than 6. ✓
5. Saying what L stands for lets you check the answer against the story and tells you which piece to find.

Balloon Message.

1.	29 − 15	= 14	N
2.	11 − 10	= 1	A
3.	2 + 11	= 13	M
4.	14 − 9	= 5	E
5.	5 × 5 − 5	= 20	T
6.	5 × 2 − 2	= 8	H
7.	6 × 2 − 7	= 5	E
8.	(14 + 9) − 2	= 21	U
9.	(10 − 3) × 2	= 14	N
10.	(6 + 9) − 4	= 11	K
11.	(14 − 12) × 7	= 14	N
12.	4 × 4 − 1 × 1	= 15	O
13.	12 × 12 − 11 × 11	= 23	W
14.	2 × 5 + 2 × 2	= 14	N

Message: **NAME THE UNKNOWN!**

Chapter 6

1. (6 × 15) → **90 minutes**, which is **1.5 hours**.
2. (5 × 298) = ((5 × 300) − (5 × 2)) = 1,500 − 10 = 1,490 cents, which is **14.90 dollars**.
3. 1.25 hours is 75 minutes. (75 + 45) → **120 minutes**, which is **2 hours**.
4. 3 days is 72 hours. 72 + 12 = **84 hours**.
5. Hours and minutes are different sizes, so 2.5 + 90 adds unlike things and gives nonsense.

Spot the Slip.
Notebook A: **perfect**. 38 × 42 = 1,596.
Notebook B: slip on line **5**. 1 × 1 is 1. The answer should be 624.
Notebook C: slip on line **3**. The corner is 4 × 4, not 4 × (4 + 1). The answer should be 2,484.

Answers (turn the book around)

Chapter 7

1. (5 × 3) → **15** outfits.
2. 15 − 2 = **13**.
3. Cut 30 into three parts of 10. Two parts: (2 × 10) → **20** stickers.
4. (6 × 7) → **42** boxes. It is a grid.
5. A table shows each outfit once and shows why the total is a product, and a rule shows up as a crossed-out box.

Color by Answer (899). The shaded boxes make a heart.



Chapter 8

1. Around 100 is friendlier: (100 × 100) − (1 × 1) = 10,000 − 1 = **9,999**.
2. Around 20: (20 × 20) − (1 × 1) = 400 − 1 = **399**.
3. (60 × 60) − (3 × 3) = 3,600 − 9 = **3,591**.
4. For example, (52 × 50) + (52 × 3) = 2,600 + 156 = **2,756**.
5. No. A bad split costs more work or more chances for a slip, but a correct redraw never changes the value. Her answer is wrong only if she made a slip along the way.

Balloon Message.

1.	5 + 14	= 19	S
2.	30 − 29	= 1	A
3.	28 − 3	= 25	Y
4.	25 − 2	= 23	W
5.	3 × 4 − 4	= 8	H
6.	5 × 2 − 9	= 1	A
7.	3 × 5 + 5	= 20	T
8.	(13 + 15) − 3	= 25	Y
9.	(13 + 9) − 7	= 15	O
10.	(12 + 12) − 3	= 21	U
11.	(6 + 7) − 2	= 11	K
12.	7 × 6 − 7 × 4	= 14	N
13.	3 × 12 − 3 × 7	= 15	O
14.	12 × 12 − 11 × 11	= 23	W

Message: **SAY WHAT YOU KNOW!**